Which Wrong Pages Should an Event Linker Train On?

Choosing the wrong examples for Event Linking

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Abstract

A news story says that a player won the Rising Star award in 1996. A reader who wants to know more needs the Wikipedia page for that award in that year, not the page for 1998. Event linking is the task of finding that page for an event mentioned in a text. A system learns the task from examples. For each example it sees the right page and a few wrong pages, and it learns to prefer the right one. Which wrong pages should it see? Pages that are obviously wrong teach it little. Pages that are almost right can mislead it.

I test four rules for choosing the wrong pages in EVELINK, the event linking system of Yu et al., and I give each rule the same amount of training. One rule uses the right pages of other examples, as EVELINK does. One takes random pages from the 1,000 that the system already ranks highest. One takes only the seven highest. The last is ReSa2, a recent rule that picks pages that score close to the right page and resemble it.

First, ReSa2 works best. The right page is among the first ten results for 93.3% of held-out mentions, compared with 92.2% for random pages from the 1,000 and 69.6% for EVELINK's rule. Second, I measure which pages each rule picks and how much its picks change from run to run. Third, I prove how little an obviously wrong page can teach, and I check the proof in Lean 4. Fourth, the seven highest pages are the worst of the three new rules, and with faster training they break the system. No rule reduces wrong-year mistakes, and none improves the results on news text.

1 Introduction

EVELINK is a system that reads a mention of an event and tries to find the Wikipedia page for that event. Yu and colleagues introduced it in 2023 [Yu et al.(2023)]. You give it a phrase like "won the Rising Star award," along with the sentence around it. EVELINK uses that context to work out which award the writer is talking about.

Take this sentence. "He won the Rising Star award in 1996, in his first season with Adelaide."

The page we want is "1996 AFL Rising Star." A page about the 2004 Indian Ocean earthquake is easy to rule out. The page for the 1998 AFL Rising Star is harder. It is the same award. The year is what tells us which page to link to. The Event Linking paper calls this mistake a time error. Its other common mistake is a hierarchy error, where the system picks a larger event that contains the right one, or a smaller event inside it. In both cases the wrong page shares most of its words with the correct page.

EVELINK makes that decision in two stages (section 2). The first stage searches all of Wikipedia

and pulls out 100 pages that might fit. It is called the *retriever*. The second stage reads the mention next to each of those 100 pages and puts them in a new order. It is called the *reranker*. If the retriever leaves out the 1996 page, the reranker cannot bring it back.

That is where this project starts. During training, the retriever sees the correct page next to a few wrong pages. I wanted to know which wrong pages help it most. A rule that picks the wrong pages is called a *sampler*, because it usually picks them at random from some set. The sampler I study is ReSa2 [Li et al.(2025)]. I ask two questions.

1. Which pages does ReSa2 pick? For each page I want the probability that it is picked, the number of different pages picked over many tries, and how much the picks change between tries.
2. With the same amount of training, does ReSa2 put the correct page higher in the results than EVELINK's own sampler does?

The paper has four results. First, I train the EVELINK retriever with four samplers, three times each, and compare them. Second, I measure what each sampler picks. Third, I prove a bound on the training loss and three counting facts about the samplers, and I check the proofs in Lean 4. Fourth, I describe a failure: with the hardest wrong pages and faster training, the retriever stops working and does not recover. The code and the logs are in the project repository.

2 Event linking: task, system, and evaluation

This section says what event linking is, how EVELINK does it, and how the result is scored.

Task. A *mention* is a short piece of a text that refers to an event, such as "won the Rising Star award." Formally, a mention $m = (d, [a, b])$ is a document d and the positions a to b of the words in it that refer to the event. Let $\mathcal{E}$ be the set of Wikipedia pages that the system can answer with. In the data of Yu et al. there are $|\mathcal{E}| = 5{,}870{,}755$ of them. Event linking is the function

$$\text{link} : m \longmapsto e_m^* \in \mathcal{E} \cup \{\text{NIL}\}, \tag{1}$$

where e_m^* is the page about the event that the mention refers to. NIL means that no page describes it. The goal is to find e_m^* for every mention from the mention and the words around it.

Data. The examples come from links that Wikipedia editors already made. When the text of an article links some words to an event page, those words are a mention and the linked page is its answer. The released data has 66,426 mentions for training, 16,692 for validation, and 19,266 for testing. Each mention is a verb ("won") or a noun ("the award"). Verbs are split by whether the same verb and the same answer appear in the training data. Nouns are split into easy and hard by how much the mention looks like the title of its page. A second set of 1,762 mentions from the New York Times tests the system on text that is not from Wikipedia. Some of those mentions have no page (NIL).

System. Reading the mention next to each of 5.87 million pages would take too long. So EVELINK works in two stages, a fast one and a careful one:

$$\text{retrieve:} \quad R_K(m) = \text{top-}\, K_{e \in \mathcal{E}}\ s(m, e), \tag{2}$$

$$\text{rerank:} \quad \hat{e}_m = \underset{e \in R_K(m)}{\arg\max}\ r(m, e). \tag{3}$$

The retriever gives every page a score $s(m, e)$ with a fast function (section 4) and keeps the $K = 100$ pages with the highest scores. The reranker reads the mention together with each of those K pages, gives each a new score $r(m, e)$ with a slow function, and returns the best page. Equation 3 is the reason this paper is about the retriever. If e^*_m is not in $R_K(m)$, the reranker never sees the right page.

Evaluation. The full system is scored by *accuracy*, the share of mentions for which it returns the right page:

$$\text{Acc} = \frac{1}{|V|} \sum_{m \in V} \mathbf{1}[\hat{e}_m = e^*_m]\,, \tag{4}$$

Here V is the set of mentions used for scoring, and $\mathbf{1}[\cdot]$ is 1 when the statement inside is true and 0 otherwise. The retriever alone is scored by *recall at* K, the share of mentions whose right page is among its first K results:

$$\text{Recall@}K = \frac{1}{|V|} \sum_{m \in V} \mathbf{1}[e^*_m \in R_K(m)]\,, \tag{5}$$

No reranker placed after the retriever can have a higher accuracy than this number. This paper changes only how the retriever is trained, so its main measure is recall at 10 on the validation set. I also report recall at 1, where the right page comes first, and recall at 100.

The released EVELINK system has an accuracy of 82.87% on the Wikipedia test set and 55.33% on the New York Times set. The released retriever alone has a recall at 10 of 77.7% on the validation set.

Where the mistakes are. Yu et al. report two kinds of mistakes. A *time error* picks the page for the same repeated event in another year, such as the 1998 Rising Star for a 1996 mention. A *hierarchy error* picks a larger event that contains the right one, or a smaller one inside it, such as the whole war for one battle. Both wrong pages share most of their words with the correct page, so the retriever gives them high scores. The retriever has to learn to rank these pages below the correct one. The samplers of section 5 differ in how often they show it such pages.

3 Related work

Event linking. Yu et al. [Yu et al.(2023)] define the task and release the Wikipedia and New York Times data. They build EVELINK from BLINK [Wu et al.(2020)], a two-stage system that links names of people and places to Wikipedia. I keep their model and their data. I change only how the wrong pages for the retriever are chosen.

Choosing wrong examples. The simplest choice is to reuse the right answers of other examples in the same batch [Karpukhin et al.(2020)]. It costs nothing, but most of these pages are easy to rule out. A harder choice is to take the pages that the retriever itself ranks highest. ANCE [Xiong et al.(2021)] does this and refreshes the pages during training. Some of the highest pages are right answers that nobody labeled. RocketQA [Qu et al.(2021)] removes those with a second model. SimANS [Zhou et al.(2022)] instead picks pages at random, and prefers pages whose score is close to the score of the right answer.

ReSa2. Li et al. [Li et al.(2025)] add a second step to SimANS. Among the pages picked in the first step, ReSa2 keeps those most similar to the right answer. I use the sizes from their released code: 1,000 pages to start, 500 after the first step, 75 after the second, and 7 used for training. Their paper tests ReSa2 on web search data. This paper measures what the two steps pick, and tests ReSa2 on event linking.

4 The retriever and its training loss

The retriever of equation 2 uses two copies of BERT, a neural network that turns a text into a list of numbers. Such a network is called an *encoder*. One encoder reads the mention and its context. The other reads the Wikipedia page. Each one turns its input into a vector with 1,024 numbers. I call the mention vector u_m and the page vector v_e.

To score a page, I take the dot product. I multiply the first number in one vector by the first number in the other, do the same for the rest, and add up the products:

$$s(m, e) = u_m \cdot v_e = \sum_{j=1}^{1024} u_{m,j}\, v_{e,j}. \tag{6}$$

The retriever puts pages with higher scores first. The two encoders work separately, so I can compute the page vectors ahead of time. When a new mention comes in, I encode it once and compare that vector with the stored page vectors.

During training, I give the retriever the correct page and seven wrong pages. I call the wrong pages *negatives*. If the correct page scores s^+ and the negatives score s_1 through s_7, the *loss*, the number that training tries to make small, is

$$L = -\log \frac{e^{s^+}}{e^{s^+} + \sum_{i=1}^{7} e^{s_i}}. \tag{7}$$

The fraction is the correct page's share of the total after I exponentiate the eight scores. When the correct page gets almost all of that share, the loss is close to zero. When a wrong page scores about as high as the correct one, it takes a bigger share and the loss goes up.

To see the difference, suppose the correct page scores 10 and there is just one negative. If that negative scores -10, the loss is about two billionths. If it scores 9.5, the loss is about 0.47. The correct page has the same score in both cases. All I changed was the wrong page.

The math puts a bound on this. Write $g_i = s^+ - s_i$ for the gap by which negative i scores below the correct page.

Lemma 1 (Easy negatives add little)**.** *For any $n \geq 1$ and real scores $s^+, s_1, \ldots, s_n$, the loss of equation 7 with n negatives satisfies*

$$L = \log\Bigl(1 + \sum_{i=1}^{n} e^{-g_i}\Bigr) \leq \sum_{i=1}^{n} e^{-g_i}, \qquad \textit{and} \qquad L \leq n\, e^{-g} \;\; \textit{if every } g_i \geq g. \tag{8}$$

Proof. Divide the numerator and the denominator of the fraction in equation 7 by e^{s^+}: $L = -\log\bigl(1/(1+\sum_i e^{s_i - s^+})\bigr) = \log(1+\sum_i e^{-g_i})$. For $x \geq 0$, $\log(1+x) \leq x$, so the first inequality holds. If every $g_i \geq g$ then each $e^{-g_i} \leq e^{-g}$, and the second follows. □

So a negative that scores g points below the correct page adds at most e^{-g} to the loss. At $g = 20$ that is about 2×10^{-9}. At $g = 0.5$ it is about 0.61. I checked Lemma 1 in Lean 4 (section 5.1).

I can also ask how much the loss changes if I raise one negative's score.

Lemma 2 (The push on a negative)**.** *The derivative of L with respect to the score of negative i is that negative's share of the total,*

$$p_i = \frac{\partial L}{\partial s_i} = \frac{e^{s_i}}{e^{s^+} + \sum_{j=1}^{n} e^{s_j}} \in (0, 1), \tag{9}$$

and $\partial L/\partial s^+ = -\sum_i p_i$.

Proof. Write $Z = e^{s^+} + \sum_j e^{s_j}$, so $L = \log Z - s^+$. Then $\partial L/\partial s_i = e^{s_i}/Z$ and $\partial L/\partial s^+ = e^{s^+}/Z - 1 = -\sum_i e^{s_i}/Z$. □

So if the retriever already puts the earthquake page far below the 1996 award, that page contributes almost nothing to the update. If the 1998 award scores close to the 1996 award, the loss puts more pressure on the model to separate them.

The update itself moves the mention vector.

Corollary 1 (The update on the mention vector)**.** *With the page vectors fixed, the gradient of L with respect to the mention vector is*

$$\nabla_{u_m} L = \sum_{i=1}^{n} p_i\,(v_{e_i} - v_{e^+}), \tag{10}$$

Proof. By the chain rule through equation 6, $\nabla_{u_m} L = \sum_i (\partial L/\partial s_i)\, v_{e_i} + (\partial L/\partial s^+)\, v_{e^+} = \sum_i p_i v_{e_i} - \bigl(\sum_i p_i\bigr) v_{e^+}$ by Lemma 2. □

Equation 10 is a weighted sum of the differences between each negative's page vector and the correct page vector. A negative that scores far below the correct page has $p_i \approx 0$ and moves nothing. A negative that scores close to it has a large p_i, and the step is along $v_{e_i} - v_{e^+}$. If the two pages are near-duplicates, that difference is small. Section 6.5 returns to this.

In the experiment, I keep the loss and the number of negatives fixed. I change which seven pages the model sees.

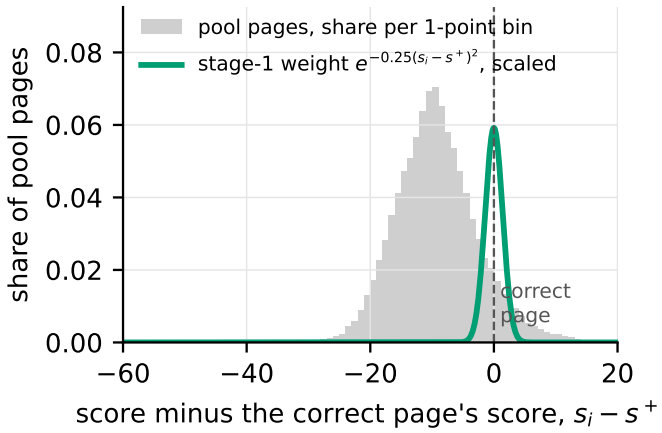


Figure 1: The weight of equation 11 (green, scaled to fit) and how far the pool pages score from the correct page (gray, 2,000 validation mentions). Most pool pages score 5 to 20 points below the correct page. The weight is close to zero for pages more than 4 points away.

5 How I chose the seven pages

I compared four samplers. The first uses the correct pages of the other seven mentions that are trained at the same time. This is EVELINK's own sampler, and it is called *in-batch*. For the other three, I started with the pages that the released retriever ranked highest for that mention. I built a pool of up to 1,000 candidates and excluded the correct page from sampling. I call these 1,000 pages the *pool*. Every sampler gets the same three things. It gets the score s_i of each pool page, the score s^+ of the correct page, and the similarity $\sigma_i = v_{e_i} \cdot v_{e^+}$ between each pool page and the correct page.

Uniform sampling gives every page in the pool the same chance, $k/N = 7/1{,}000$. *Top-score* sampling always takes the seven pages with the highest scores. Those seven are picked with probability 1 and the rest with probability 0. ReSa2 makes the choice in two stages.

First, ReSa2 draws 500 pages without replacement. It gives the most weight to pages that score close to the correct page:

$$w_i = \exp\!\left(-\tau\,(s_i - s^+)^2\right), \qquad \tau = 0.25. \tag{11}$$

It draws one page with probability $w_i / \sum_j w_j$, removes it, and draws again from the pages that are left. The weight drops as the score moves away from s^+ in either direction. Figure 1 shows the weight next to the real scores. The scores are about 550, and most pool pages are 5 to 20 points below the correct page. The weight is close to zero for any page more than 4 points away. So stage 1 first takes the few pages near s^+, and then fills up to 500 with nearly random pages from the rest. Then ReSa2 compares the 500 pages that are left with the correct page itself. It keeps the 75 pages whose vectors are most similar to the correct page vector and draws seven of those uniformly.

That last draw gives a simple bound. Once a page makes the final 75, it has a 7-in-75 chance of being used. Suppose a page is labeled as wrong but is in fact a valid answer. Then I can bound how often ReSa2 trains against it. With equation 9, I can also bound its expected push on the

model:

$$\begin{aligned}\mathbb{E}[p_i] &\le P(\text{ReSa2 picks } i) \cdot \frac{e^{s_i}}{e^{s^+} + e^{s_i}} \\ &\le \frac{7}{75} \cdot \frac{e^{s_i}}{e^{s^+} + e^{s_i}}.\end{aligned} \tag{12}$$

5.1 What is proved in Lean

Lean is a programming language in which a proof is a program, and the computer checks every step. I wrote four of the claims in this paper in Lean 4. The files are in Appendix B. The first claim is the loss bound of equation 8. For the other three I need one definition. A *possible false negative* is a pool page that scores above the correct page, $s_i > s^+$. Such a page may be a valid answer that nobody labeled.

1. Lemma 1, for any number of negatives with real scores. The Lean names are `loss_eq_log_gaps`, `loss_le_sum_exp_neg_gap`, and `loss_le_of_gap_ge`. This proof uses Mathlib, the Lean library of mathematics.
2. Proposition 1 (`any_sampler_le`) and Proposition 2 (`topscore_is_max`).
3. Proposition 3 (`stage2_share`, checked by computation).

Proposition 1 (A bound for every sampler)**.** *Let the pool contain A possible false negatives. Any selection of k distinct pages from the pool contains at most* $\min(k, A)$ *of them.*

Proof. The selection has k pages, and it is a sub-list of the pool, so it cannot contain more pages above s^+ than the pool does. □

Proposition 2 (Top-score reaches the bound)**.** *The top-score rule selects exactly* $\min(k, A)$ *possible false negatives.*

Proof. Sorting by score puts every page above s^+ before every page below it, so the first k pages are the $\min(k, A)$ pages above s^+ followed, if $k > A$, by $k - A$ pages below it. □

Together, Propositions 1 and 2 say that no sampler trains against more possible false negatives than top-score does. This holds for every pool and every draw, and it needs no assumption about ReSa2.

Proposition 3 (The 7/75 share)**.** *Of all 7-page subsets of a set of 75 pages, the fraction that contains one fixed page is* $7/75$*; equivalently,* $75\binom{74}{6} = 7\binom{75}{7}$*. Hence* $P(\textit{ReSa2 selects } i) = \frac{7}{75}\, P(i \textit{ is among the 75 kept}$ *and equation 12 follows.*

Proof. A subset that contains the fixed page is the fixed page plus 6 of the other 74 pages, so there are $\binom{74}{6}$ of them out of $\binom{75}{7}$, and $\binom{74}{6}/\binom{75}{7} = 7/75$. The identity is checked by computation in Lean. □

The statements as Lean reads them are in Appendix B.

Three things are not in Lean. Lemma 2 and Corollary 1 are proved by hand above. For stage 1 of ReSa2, I computed the exact probability that each page is picked in pools of up to 8 pages, and it agrees with a simulation of my code to within 0.015. The results of section 6 are measurements.

Table 1: Recall on the 16,692 validation mentions after 1,500 updates. Each cell is the mean of three runs ± their standard deviation. "Released" is the model before my training. The three runs are listed one by one in Appendix A.

Sampler	Top-1	R@10	R@100
Released	42.6	77.7	93.1
In-batch	33.6 ± 1.9	69.6 ± 1.8	89.8 ± 1.3
Uniform	64.3 ± 0.2	92.2 ± 0.1	98.4 ± 0.1
Top-score	65.3 ± 1.2	87.2 ± 0.9	95.9 ± 0.6
ReSa2	**69.5** ± 0.3	**93.3** ± 0.1	**98.5** ± 0.1

6 Experiments

6.1 Setup

I started each run from the released EVELINK model. I kept the page encoder fixed and trained only the mention encoder. Since the page vectors never change, the pool of each mention and its scores can be computed once, before training.

Each run made 1,500 updates. One update uses 32 mentions, each with seven negatives. The learning rate, which sets the size of each update, was 10^{-5}. It rises from zero over the first 10% of the updates. The other settings are those of EVELINK. Training has some randomness, so I ran each of the four samplers three times with different random seeds. That gives twelve runs. Each run took about 56 minutes on one NVIDIA A10G GPU.

Then I ranked all 5.87 million pages for each of the 16,692 validation mentions and computed recall at 1, 10, and 100 (equation 5). The measure I focus on is recall at ten. If the correct page appears anywhere in the first ten results, I count that mention as a hit.

Before the experiments I checked that I could repeat the published results. The released retriever and reranker gave me an accuracy of 82.88% on the Wikipedia test set, and the paper reports 82.87%. On the New York Times set I got 55.33%, the same as the paper. I did not use the test set for anything else.

6.2 What changed in the rankings

Table 1 gives the results. The released retriever got 77.7 percent recall at ten. After fine-tuning, in-batch sampling got 69.6. Uniform sampling from the pool got 92.2. Taking the seven highest-scoring negatives got 87.2. ReSa2 got 93.3.

Most of the improvement came from drawing negatives from the pool. ReSa2 added 1.1 percentage points over uniform sampling at ten. The difference was larger at rank one. With uniform sampling, the correct page came first 64.3 percent of the time. With ReSa2, it came first 69.5 percent of the time. For both samplers the three runs differ by 0.1 to 0.3 points, so the gap between them is much larger than the differences between runs.

In-batch sampling finished below the released model. I think the reason is the small number of negatives. Seven random pages are almost always easy, and by Lemma 1 easy pages add almost nothing to the loss. The original training of EVELINK used many more pages at each update and ran for longer.

Table 2: What each sampler picks, over 500 validation mentions and 200 draws per mention. The first column splits the seven negatives of one draw by how similar they are to the correct page: among the 25 most similar pool pages, the next 50, the next 225, or the rest. The overlap of two draws is the number of pages they share divided by the number of pages in either.

Sampler	Negatives by similarity rank top 25 / 25–75 / 75–300 / rest	Distinct pages (of 1,000)	Overlap of two draws	Score gap to correct page
Uniform	0.17 / 0.34 / 1.55 / 4.95	756	0.00	−13.3
Top-score	4.13 / 1.07 / 1.17 / 0.62	7	1.00	−1.4
ReSa2	2.15 / 3.98 / 0.87 / 0.00	84	0.05	−8.6

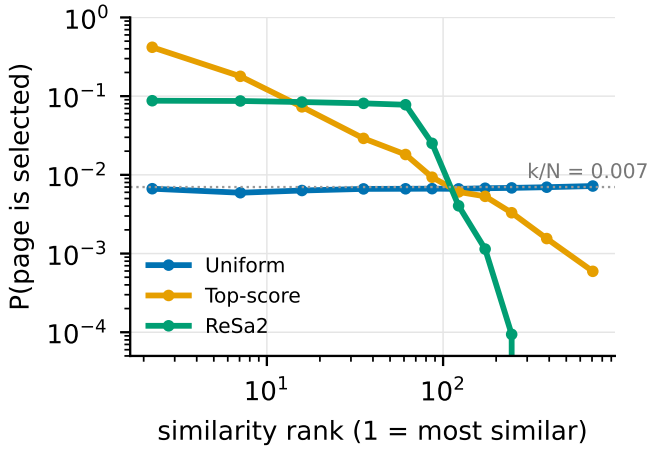


Figure 2: The probability that a pool page is picked, by how similar it is to the correct page. Rank 1 is the most similar page. Both axes are logarithmic. Uniform gives every page the same chance. ReSa2 gives about 0.09 to each of the 75 most similar pages and picks nothing beyond rank 300.

6.3 Which pages the samplers pick

I also wanted to see which pages the samplers actually picked. I drew negatives 200 times each for 500 validation mentions. Table 2 and Figure 2 give the result.

ReSa2 picked about six of its seven negatives from the 75 pages most similar to the correct page. For uniform sampling, the average was about half a page. Top-score gave me the same seven pages every time. Two ReSa2 draws shared very few pages, even though both tended to draw from the same part of the pool. Over 200 draws, uniform sampling used 756 different pool pages. ReSa2 used 84, and top-score used 7.

The last column says how hard the negatives are. On average, ReSa2's negatives score 8.6 points below the correct page. Uniform negatives score 13.3 points below it, so they are easier. Top-score negatives score only 1.4 points below it, so they are the hardest. Figure 2 shows the same picture page by page. Top-score puts most of its picks on the first few pages, which are the near-copies of the correct page.

Table 3: Recall at ten on the validation set by kind of mention (mean of three runs ± standard deviation). From left to right the columns hold 1,814, 2,585, 3,947, 4,173, and 4,173 mentions.

	Verb			Noun	
Sampler	Seen	Unseen form	Unseen	Hard	Easy
Released	76.1	77.3	82.9	68.9	82.6
In-batch	65.2 ± 1.5	69.0 ± 1.3	75.0 ± 0.6	58.0 ± 1.5	78.4 ± 3.9
Uniform	98.8 ± 0.2	91.3 ± 0.2	88.3 ± 0.1	88.8 ± 0.2	97.1 ± 0.3
Top-score	96.5 ± 0.2	83.2 ± 1.5	79.6 ± 1.2	84.9 ± 1.0	95.2 ± 0.5
ReSa2	**99.1** ± 0.0	**92.3** ± 0.1	**90.2** ± 0.1	**90.3** ± 0.2	**97.5** ± 0.2

Table 4: Mentions whose first result is wrong, as a percentage of all 16,692 validation mentions, split by kind of mistake.

Sampler	Wrong first result	Time	Hierarchy	Other
Released	57.4	1.8	11.6	44.0
In-batch	66.4 ± 1.9	1.9 ± 0.1	11.7 ± 0.6	52.8 ± 1.4
Uniform	35.7 ± 0.2	2.1 ± 0.1	5.5 ± 0.2	28.1 ± 0.2
Top-score	34.7 ± 1.2	2.1 ± 0.1	3.0 ± 0.1	29.6 ± 1.2
ReSa2	30.5 ± 0.3	2.0 ± 0.1	4.4 ± 0.2	24.0 ± 0.1

6.4 Mention classes, error types, and another domain

I looked at the same twelve runs in three more ways. For this I scored the released retriever and the twelve trained ones again, and I kept the position of the correct page for every mention.

By kind of mention. Table 3 splits recall at ten by the kinds of mention from section 2. ReSa2 is the best sampler for every kind. The released model was weakest on hard nouns, at 68.9. ReSa2 brings them to 90.3, and uniform sampling to 88.8. ReSa2's largest lead over uniform is on unseen verbs, 90.2 compared with 88.3. These are the verbs whose answer never appears in the training data.

By kind of mistake. For every validation mention whose first result is wrong, I compared the title of that result with the title of the correct page. I count a *time error* when the two titles are the same after I remove the years. I count a *hierarchy error* when all the words of one title appear in the other title, and the other title has more words. This rule reads only titles, so it misses some mistakes of both kinds. It is the same rule for every model. Table 4 gives the result.

No sampler changes the time errors. They stay at about 2% of mentions for every model. I expected this risk in my proposal. When the model makes a time error, the wrong-year page scores above the correct page. Stage 1 of ReSa2 gives little weight to pages that score well above the correct page, so the retriever rarely trains against them.

Hierarchy errors do fall. Top-score sampling leaves 3.0%, ReSa2 leaves 4.4%, and uniform leaves 5.5%. The samplers that show the retriever more of these close pages leave fewer of these mistakes. ReSa2 wins overall because it leaves the fewest mistakes of the third kind.

Table 5: Recall on the 769 New York Times mentions that have a Wikipedia page (mean of three runs ± standard deviation).

Sampler	Top-1	Recall@10	Recall@100
Released	**13.0**	**33.6**	**57.3**
In-batch	8.5 ± 1.1	24.4 ± 1.4	44.8 ± 1.7
Uniform	9.8 ± 0.7	27.4 ± 1.0	50.8 ± 1.6
Top-score	4.6 ± 0.7	14.4 ± 1.4	26.8 ± 1.8
ReSa2	11.5 ± 0.1	30.9 ± 1.5	55.9 ± 0.4

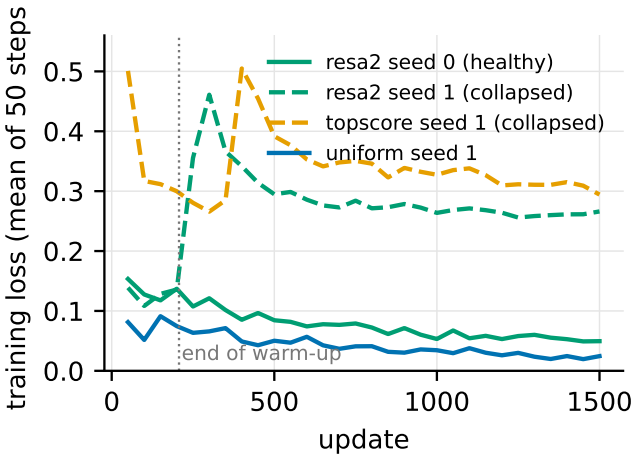


Figure 3: Training loss at the higher learning rate for two healthy runs (solid) and the two collapsed runs (dashed). The dotted line marks the update at which the learning rate reaches its highest value. The collapsed runs jump once and stay at about 0.3.

On news text. Table 5 gives the same retrievers on the 769 New York Times mentions that have a page. Here my training made things worse. No trained retriever reaches the released model's recall at ten of 33.6. ReSa2 loses the least and ends at 30.9. Top-score loses the most and ends at 14.4. The order of the samplers is the same as on Wikipedia. So the choice of negatives matters here too, but 1,500 updates on Wikipedia text do not make a better retriever for news.

6.5 What happened at the higher learning rate

My first twelve runs used a learning rate three times higher, 3×10^{-5}. One ReSa2 run and one top-score run collapsed. Figure 3 shows the training loss. It jumped once, just after the learning rate reached its highest value, from about 0.13 to about 0.3, and it stayed there. When I looked at the scores, I found that the mention encoder gave almost every page a score of about two. It gave roughly the same score to the correct page and to a random page. A healthy model gives about 550 to the correct page and 545 to a random one. The weights stayed finite. The loss over the eight training pages could still look modest while the ranking across millions of pages had fallen apart.

One possible explanation is in Corollary 1. When a negative scores close to the correct page, it gets a substantial share of the training signal. If the two page vectors are also similar, the update depends on the small difference between them. My hypothesis is that repeated updates at the higher learning rate weakened the part of the mention vector that lined up with the page vectors. The scores are consistent with that explanation. Uniform and in-batch runs, whose negatives are

far from the correct page, never jumped.

I tried training the collapsed model for another 1,500 updates. I tried ReSa2 negatives in one run and uniform negatives in another. Recall at ten stayed at zero in both. Then I started again from the released model with a learning rate of 10^{-5}. All twelve runs trained without collapsing, and every sampler scored better than at the higher rate (Appendix A). Those twelve runs are the ones reported above.

7 Conclusion

I started with the difference between the 1996 award and the 1998 award. The results make that example more specific for me. I want the retriever to see pages that are close enough to make it pay attention to the year. I also want it to see a different set of pages the next time it trains on that mention. In these runs, ReSa2 gave me that variety and put the correct page first more often than any of the other samplers I tested. It did not fix everything. The wrong-year errors that started this paper are as frequent after training as before, with every sampler, and on news text the released model still retrieves better than any of the fine-tuned ones.

Limitations

All Wikipedia numbers are on the validation set. I did not use the test set. The kinds of mistake in section 6.4 come from a rule on page titles, not from a person who read each case.

The page encoder is frozen, so the pool scores and similarities never change during training. A run that also trains the page encoder could behave differently, and its pools would have to be rebuilt as the encoder moves.

Each sampler ran with one amount of training (1,500 updates, 7 negatives) and two learning rates. The collapse appeared at one of them. My explanation for it is a guess that fits the scores. I did not prove it.

The pool is the released retriever's top 1,000 pages. A correct page outside the pool (2% of mentions) can never be sampled.

The Lean results cover the loss bound and the counting facts about possible false negatives. The derivative and the gradient are checked by hand.

How I used AI

The course asks for a description of how AI tools were used. I used two. Before the proposal, I used Codex to search for papers from researchers in the instructor's academic circle about linking documents, which led to Event Linking, and then for a more recent negative sampler, which led to ReSa2 and SimANS. It also wrote "learning versions" of the papers that explained them through the concepts of this course (conditional probability, indicator variables, expectation) with small examples; I read those first and the full papers second.

During the project I used Claude, through an agent that ran on my computer and on a cloud GPU. It set up the cloud account and the GPU machine. It changed the released EVELINK code so that the model fits a 24 GB GPU (a low-memory candidate store and mixed precision). From my

descriptions it wrote the ReSa2 sampler and the hook that lets the training loop swap samplers, and it wrote the Lean proofs. It ran the 24 training runs and the recovery test, built the tables, and drew the figures. Along the way it fixed its own mistakes: an out-of-memory crash, a full disk, and a bug in the original validation code. The words are mine. I narrated each section, the agent turned the narration into plain technical English, and I corrected the result. The repository has an AI-usage log with one line per session, and I kept the prompts and outputs; I can produce the key ones on request.

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A Per-seed results

Each cell gives recall at 1, 10, and 100 on the validation set for one run. The runs at learning rate 10^{-5} are the main results (Table 1). The runs at 3×10^{-5} are my first attempt, in which two runs collapsed.

The recovery test continued the collapsed top-score seed-1 model for 1,500 more updates: with ReSa2 negatives it scored 0.0 / 0.0 / 2.2, with uniform negatives 0.0 / 0.0 / 1.1.

Table 6: Learning rate 10^{-5}.

Sampler	Seed 0	Seed 1	Seed 2
In-batch	36.0 / 71.7 / 91.2	31.4 / 67.4 / 88.1	33.4 / 69.7 / 90.1
Uniform	64.4 / 92.3 / 98.5	64.5 / 92.4 / 98.4	64.1 / 92.0 / 98.4
Top-score	64.4 / 86.5 / 95.6	64.6 / 86.7 / 95.4	67.0 / 88.5 / 96.7
ReSa2	70.0 / 93.2 / 98.4	69.4 / 93.3 / 98.5	69.3 / 93.4 / 98.6

Table 7: Learning rate 3×10^{-5} (first attempt).

Sampler	Seed 0	Seed 1	Seed 2
In-batch	24.4 / 57.6 / 82.9	22.7 / 56.4 / 83.1	25.0 / 60.7 / 85.0
Uniform	60.7 / 90.0 / 98.0	61.5 / 90.3 / 97.9	61.1 / 90.4 / 97.8
Top-score	55.2 / 78.3 / 90.1	0.0 / 0.1 / 0.9	57.8 / 80.6 / 92.3
ReSa2	68.3 / 92.0 / 98.0	0.0 / 0.1 / 1.2	67.5 / 92.1 / 98.1

B The Lean file

Two files in the project repository, Lean 4 v4.34.1, built with `lake build`. `lean/LossBound.lean` uses Mathlib (real numbers, exp, log); its proofs depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. `lean/FalseNegatives.lean` uses the core library only.

LossBound.lean

```
import Mathlib.Analysis.SpecialFunctions.Log.Basic
import Mathlib.Algebra.BigOperators.Fin

/-!
# Easy negatives add almost nothing to the loss

Uses Mathlib for the real numbers, `Real.exp`, and `Real.log`. Check with:
`cd lean && lake build`.

Setup. One training mention has a correct page with score `sPos` and `n`
negatives with scores `s i`. The retriever loss (softmax cross entropy) is

  L = -log (e^{sPos} / (e^{sPos} +  e^{s i})).

The gap of negative `i` is `g i = sPos - s i`: how far it scores below the
correct page.

Results.
* `loss_eq_log_gaps`: L = log (1 +  e^{-g i}). The loss depends only on
  the gaps.
* `loss_le_sum_exp_neg_gap`: L ≤  e^{-g i}, because log (1 + x) ≤ x.
  So a negative with gap g adds at most e^{-g} to the bound.
* `loss_nonneg`: L ≥ 0.
-/

open Real Finset

/-- The softmax cross-entropy loss of one mention: correct-page score `sPos`,
negative scores `s 0, ..., s (n-1)`. -/
noncomputable def loss {n : ℕ} (sPos : ℝ) (s : Fin n → ℝ) : ℝ :=
  -Real.log (Real.exp sPos / (Real.exp sPos + ∑ i, Real.exp (s i)))

/-- The loss written with the gaps `sPos - s i`. -/
theorem loss_eq_log_gaps {n : ℕ} (sPos : ℝ) (s : Fin n → ℝ) :
```

```
    loss sPos s = Real.log (1 + ∑ i, Real.exp (-(sPos - s i))) := by
  have hpos : 0 < Real.exp sPos := Real.exp_pos sPos
  have hsum : 0 ≤ ∑ i, Real.exp (s i) :=
    Finset.sum_nonneg fun i _ => (Real.exp_pos (s i)).le
  -- e^{-(sPos - s i)} = e^{s i} / e^{sPos}
  have hgap : ∀ i, Real.exp (-(sPos - s i)) = Real.exp (s i) / Real.exp sPos := by
    intro i
    rw [neg_sub, Real.exp_sub]
  -- 1 +  e^{-g i} = (e^{sPos} +  e^{s i}) / e^{sPos}
  have hrw : 1 + ∑ i, Real.exp (-(sPos - s i))
      = (Real.exp sPos + ∑ i, Real.exp (s i)) / Real.exp sPos := by
    rw [eq_div_iff hpos.ne', add_mul, one_mul, Finset.sum_mul]
    congr 1
    refine Finset.sum_congr rfl fun i _ => ?_
    rw [hgap i, div_mul_cancel₀ _ hpos.ne']
  rw [hrw, loss,  Real.log_inv, inv_div]

/-- The loss is never negative. -/
theorem loss_nonneg {n : ℕ} (sPos : ℝ) (s : Fin n → ℝ) : 0 ≤ loss sPos s := by
  rw [loss_eq_log_gaps]
  apply Real.log_nonneg
  have : 0 ≤ ∑ i, Real.exp (-(sPos - s i)) :=
    Finset.sum_nonneg fun i _ => (Real.exp_pos _).le
  linarith

/-- Main bound: L ≤  e^{-g i}. Each negative adds at most e^{-g} to it. -/
theorem loss_le_sum_exp_neg_gap {n : ℕ} (sPos : ℝ) (s : Fin n → ℝ) :
    loss sPos s ≤ ∑ i, Real.exp (-(sPos - s i)) := by
  rw [loss_eq_log_gaps]
  set x := ∑ i, Real.exp (-(sPos - s i)) with hx
  have hx0 : 0 ≤ x := Finset.sum_nonneg fun i _ => (Real.exp_pos _).le
  -- log y ≤ y - 1 for y > 0, with y = 1 + x
  have := Real.log_le_sub_one_of_pos (by linarith : (0 : ℝ) < 1 + x)
  linarith

/-- Corollary: if every negative scores at least `g` below the correct page,
then L ≤ n · e^{-g}. -/
theorem loss_le_of_gap_ge {n : ℕ} (sPos g : ℝ) (s : Fin n → ℝ)
    (hg : ∀ i, g ≤ sPos - s i) :
    loss sPos s ≤ n * Real.exp (-g) := by
  calc loss sPos s ≤ ∑ i, Real.exp (-(sPos - s i)) := loss_le_sum_exp_neg_gap sPos s
    _ ≤ ∑ _i : Fin n, Real.exp (-g) :=
        Finset.sum_le_sum fun i _ => Real.exp_le_exp.mpr (by linarith [hg i])
    _ = n * Real.exp (-g) := by simp
```

FalseNegatives.lean

The opening comment and the three examples at the end of the file are left out here.

```
/-- `s` scores above the correct page. -/
def above (sPos s : Int) : Bool := decide (sPos < s)

/-- Number of possible false negatives in a list of pages. -/
def nAbove (sPos : Int) (l : List Int) : Nat := l.countP (above sPos)

/-! ## 1. An upper bound for every sampler -/

/-- Any selection taken from the pool (each page used at most once) has at most
`min (its size) A` possible false negatives. (Write the selected pages in
pool order; then "each page used at most once" means `sel` is a sublist.) -/
theorem any_sampler_le (sPos : Int) (pool sel : List Int) (h : sel.Sublist pool) :
    nAbove sPos sel ≤ min sel.length (nAbove sPos pool) :=
  Nat.le_min.mpr ⟨List.countP_le_length, h.countP_le⟩

/-! ## 2. The top-score rule reaches the bound
```

```
Sorting by score puts every page above ‘sPos‘ before every page below it.
So, for counting, top-score is: pages above first, then the others, take ‘k‘. -/

def topscore (sPos : Int) (k : Nat) (pool : List Int) : List Int :=
  (pool.filter (above sPos) ++ pool.filter (fun s => !above sPos s)).take k

/-- Taking ‘k‘ from (all-good list ++ no-good list) gives ‘min k (good length)‘ goods. -/
theorem countP_take_append (p : Int → Bool) (a b : List Int)
    (ha : ∀ x ∈ a, p x = true) (hb : ∀ x ∈ b, p x = false) :
    ∀ k, ((a ++ b).take k).countP p = min k a.length := by
  induction a with
  | nil =>
    intro k
    simp only [List.nil_append, List.length_nil, Nat.min_zero]
    exact List.countP_eq_zero.mpr fun x hx => by
      simp [hb x (List.mem_of_mem_take hx)]
  | cons x xs ih =>
    intro k
    cases k with
    | zero => simp
    | succ k =>
      have hx : p x = true := ha x (by simp)
      have ih' := ih (fun y hy => ha y (by simp [hy])) k
      simp only [List.cons_append, List.take_succ_cons, List.countP_cons, hx,
        ite_true, List.length_cons, ih']
      omega

theorem topscore_is_max (sPos : Int) (k : Nat) (pool : List Int) :
    nAbove sPos (topscore sPos k pool) = min k (nAbove sPos pool) := by
  unfold nAbove topscore
  rw [countP_take_append (above sPos) _ _ (fun x hx => (List.mem_filter.mp hx).2)
      (fun x hx => by simpa using (List.mem_filter.mp hx).2)]
  rw [List.countP_eq_length_filter]

/-! ## 3. Stage 2 of ReSa2 picks 7 of the top 75 uniformly

Counting fact: of all 7-page subsets of 75 pages, the share that contains one
fixed page is 7/75. So P(page selected) = (7/75) · P(page is in the top 75),
which is at most (7/75) · P(page is kept by stage 1). -/

/-- Binomial coefficient. Step i: C(n, i+1) = C(n, i) · (n − i) / (i + 1),
    and each division is exact. -/
def choose (n k : Nat) : Nat :=
  (List.range k).foldl (fun c i => c * (n - i) / (i + 1)) 1

/-- Subsets that contain the fixed page: choose the other 6 from 74.
    ‘75 · C(74,6) = 7 · C(75,7)‘, that is, the share is 7/75. -/
theorem stage2_share : 75 * choose 74 6 = 7 * choose 75 7 := by decide
```